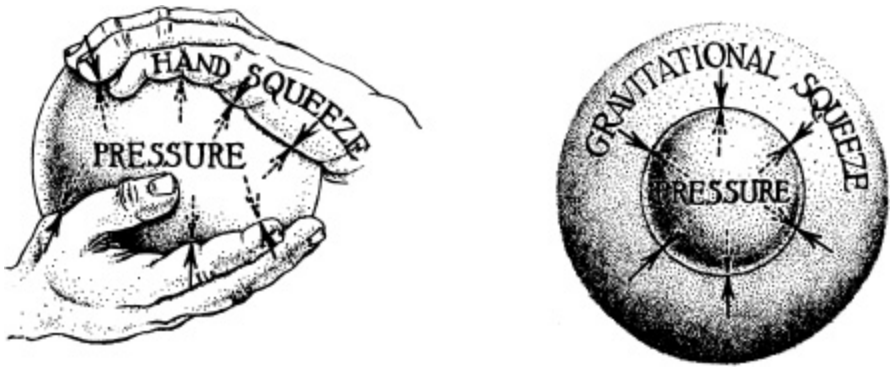


4.2 *Left:* The balance between the squeeze of your hands and the pressure inside a balloon. *Right:* The analogous balance between the gravitational squeeze (weight) of an outer shell of stellar matter and the pressure of an inner ball of stellar matter.

For a star (right half of Figure 4.2) the analogue of your squeezing hands is the weight of an outer shell of stellar matter, and the analogue of the air in the balloon is the spherical ball of matter inside that shell. The boundary between the outer shell and inner ball can be chosen anywhere one wishes—a meter deep into the star, a kilometer deep, a thousand kilometers deep. Wherever one chooses the boundary, it must fulfill the requirement that the weight of the outer shell squeezing on the inner ball (the outer shell's "gravitational squeeze") is precisely counterbalanced by the pressure of the inner ball's molecules bombarding the outer shell. This balance, enforced at each and every location inside the star, determines the star's *structure*,. that is, it determines the details of how the star's pressure, gravity, and density vary from the star's surface down to its center.

Eddington's book also described a troubling paradox in what was then known about the structures of white-dwarf stars. Eddington believed—indeed all astronomers believed in 1925—that the pressure of white-dwarf matter, like that in your balloon, must be caused by its heat. Heat makes the matter's atoms fly about inside the star at high speed, bombarding each other and bombarding the interface between the star's outer shell and its inner ball. If we take a "macroscopic" viewpoint, too coarse to detect the individual atoms, then all we can measure is the total bombardment force of all the atoms that hit, say, one square centimeter of the interface. That total force is the star's pressure.

As the star cools by emitting radiation into space, its atoms will fly about more slowly, their pressure will go down, and the weight of the star's outer shell will then squeeze its inner ball into a smaller volume. This compression of the ball, however, heats it up again, raising its pressure so a